

Fortgeschrittenenpraktikum

Raman Spectroscopy

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Abstract

The vibration Raman effect consists of the appearance of vibration inelastic (Stokes) and superelastic (Antistokes) lines in the fluorescence spectrum of gases, liquids and solids. In this experiment the vibration Raman spectra of various liquid substances are recorded and identified.

1 Theoretical background

1.1 Raman scattering

Vibrational spectroscopies are employed in the determination of the molecular structures and are used in many areas. Among the available techniques, Raman spectroscopy is essential in the study of nanostructured materials and biological systems. Detailed information of the molecular structure can be obtained, because each vibration of atoms shows a characteristic position and intensity. Both position and intensity are influenced by the chemical bonds, inter- and intramolecular forces [1]

An inelastic scattering process is responsible for the appearance of vibrational bands (Raman spectra). Unlike in IR spectroscopy (where the photon energy must coincide with the energy difference between two available vibrational states), in Raman spectroscopy a molecule can scatter a monochromatic radiation. The Raman spectrum appears in a wavelength λ slightly higher or lower than the incident radiation [1].

Thus Raman scattering can be visualized as a perturbation of the molecule by incident photon, which eigenstates can be described as a linear combination of the stationary states (or time-independent) wave functions of the molecule. The photon released when the molecule returns to electronic ground state has, more often, the same incident photon energy (Rayleigh scattering), but Raman scattering (which contains the vibrational information) can also be present. [1]

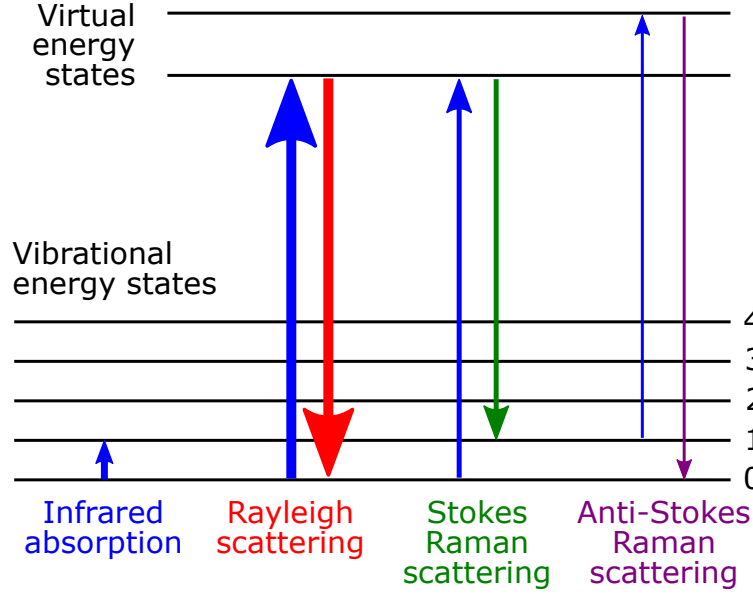


Figure 1: Energy-level diagram showing the states involved in Raman spectra [2]

A vibration (vibration coordinate ρ) is *Raman active* when the polarizability α (or *induced dipolar momentum*) of the molecule is changed (described as $\frac{\partial \alpha}{\partial \rho} \neq \emptyset$). This illustrates that the nature of the Raman effect is physically different from the infrared. Selection rules are also different and thus lead to a different spectra [1].

1.2 Lock-in amplifiers

A lock-in amplifier is a type of amplifier that extracts a signal with a known carrier wave from a noisy environment. Depending on the instrument, signals up to orders of magnitude smaller than the noise components, potentially close by in frequency, can still be reliably detected [3]. This reliability depends on the *dynamic reserve* of the instrument, which is simply a measure of how well the instrument can extract the signal when there's a much larger unwanted noise sitting on top of it. Modern instruments achieve a dynamic reserve up to 120 dB [3].

For amplitude quantities,

$$\text{dB} = 20 \log_{10} \left(\frac{A_{\text{noise}}}{A_{\text{signal}}} \right) \quad (1)$$

So for 120 dB, the above equation yields

$$120 \text{ dB} = 20 \log_{10} \left(\frac{A_{\text{noise}}}{A_{\text{signal}}} \right) \quad (2)$$

which gives

$$\frac{A_{\text{noise}}}{A_{\text{signal}}} = 10^{\frac{120}{20}} = 10^6 \quad (3)$$

So modern instruments can theoretically recover signals with an order of magnitude 10^{-6} smaller than the noise.

A lock-in amplifier performs a multiplication of its input with a reference signal, also sometimes called down-mixing or heterodyne/homodyne detection, and then applies an adjustable low-pass filter to the result [4]. This method is termed demodulation or phase-sensitive detection and isolates the signal at the frequency of interest from all other frequency components. The reference signal is either generated by the lock-in amplifier itself or provided to the lock-in amplifier and

the experiment by an external source. The reference signal is usually a sine wave but could have other forms, too. Demodulation with a pure sine wave enables selective measurement at the fundamental frequency or any of its harmonics [4].

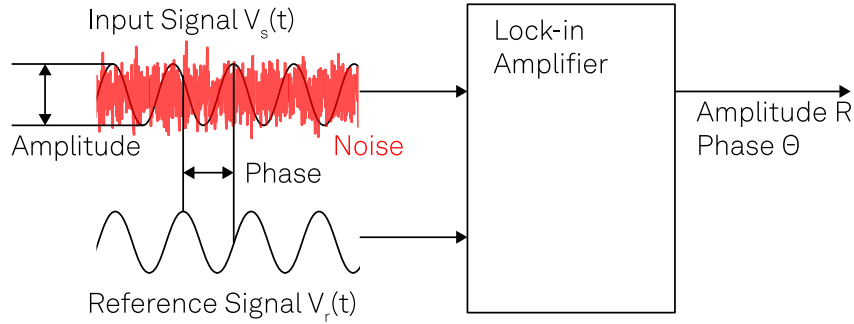


Figure 2: Lock-in amplifiers are capable of measuring the amplitude and the phase of a signal relative to a defined reference signal, even if the signal is entirely buried in noise. Adapted from [4]

1.3 Classical and Quantum Mechanical Theory

The classical picture of Raman scattering starts with the electric field of the incident laser light,

$$E(t) = E_0 \cos(\omega_0 t), \quad (4)$$

which induces a dipole moment in the molecule,

$$p(t) = \alpha E(t), \quad (5)$$

where α is the molecular polarisability.

Since the molecule is vibrating, its polarisability can change with the normal coordinate Q of the vibration:

$$\alpha(Q) = \alpha_0 + \left(\frac{\partial \alpha}{\partial Q} \right)_0 Q. \quad (6)$$

If the vibration has frequency ω_v , the induced dipole contains components at

$$\omega_0, \quad \omega_0 - \omega_v, \quad \omega_0 + \omega_v. \quad (7)$$

These correspond to:

- **Rayleigh scattering:** ω_0
- **Stokes Raman scattering:** $\omega_0 - \omega_v$
- **anti-Stokes Raman scattering:** $\omega_0 + \omega_v$

In the quantum mechanical picture, the incident photon interacts with the molecule through a virtual state. The molecule changes its vibrational energy, and therefore the scattered photon has a different energy:

$$E_{\text{out}} = E_{\text{in}} \mp \hbar \omega_v. \quad (8)$$

Thus, the Raman shift directly provides information about the vibrational energy levels of the molecule.

1.4 Polarisability

Polarisability describes how easily the electron cloud of a molecule can be distorted by an external electric field. The induced dipole moment is given by

$$\mathbf{p}_{\text{ind}} = \boldsymbol{\alpha}\mathbf{E}, \quad (9)$$

where $\boldsymbol{\alpha}$ is generally a tensor.

The important Raman selection rule is

$$\left(\frac{\partial \alpha}{\partial Q} \right)_0 \neq 0 \quad (10)$$

A vibrational mode is therefore **Raman active if the vibration produces a change in the molecular polarisability**.

1.5 Degree of Depolarisation

The degree of depolarisation describes how much the polarisation of the scattered Raman light differs from that of the incident laser light. It is commonly defined as

$$\rho = \frac{I_{\perp}}{I_{\parallel}}, \quad (11)$$

where $I_{\perp}$ is the intensity of the Raman light polarised perpendicular to the incident polarisation and $I_{\parallel}$ is the intensity parallel to it.

For the usual treatment of Raman scattering from randomly oriented molecules,

$$0 \leq \rho \leq \frac{3}{4}. \quad (12)$$

A low value of ρ corresponds to strongly polarised Raman scattering, whereas a larger value indicates stronger depolarisation. The degree of depolarisation can provide information about the symmetry of a vibrational mode.

1.6 Scattering Cross Section

The scattering cross section σ describes the probability that an incident photon will be scattered by a molecule. It can be interpreted as an effective area that characterises the scattering probability.

Raman scattering has a very small scattering cross section compared with Rayleigh scattering. Consequently, Raman signals are weak and require sensitive detectors and sufficiently strong laser illumination.

1.7 Normal Coordinates and Normal Oscillations

A molecule can undergo many different vibrational motions. These motions can be decomposed into independent **normal modes**.

Each normal mode has:

- a characteristic pattern of atomic motion,
- a characteristic vibrational frequency, and
- a normal coordinate Q describing the amplitude of the vibration.

The molecular motion can therefore be described using a set of normal coordinates,

$$Q_1, Q_2, Q_3, \dots \quad (13)$$

which correspond to the different normal modes of vibration.

The dependence of the polarisability on a normal coordinate is particularly important for Raman spectroscopy:

$$\alpha(Q) = \alpha_0 + \left(\frac{\partial \alpha}{\partial Q} \right)_0 Q + \dots \quad (14)$$

1.8 Induced and Permanent Dipole Moment

A molecule may possess a **permanent dipole moment** due to an asymmetric distribution of electric charge,

$$\mathbf{p}_0 \neq 0. \quad (15)$$

An external electric field can also distort the electron cloud and produce an **induced dipole moment**:

$$\mathbf{p}_{\text{ind}} = \alpha \mathbf{E}. \quad (16)$$

Raman scattering originates from this induced dipole. The electric field of the laser drives the induced dipole, which subsequently radiates the scattered light.

1.9 Raman shift wavenumber

Given that energy shifts in the incident photons are not wavelength dependent, it makes sense to express the raman spectra in a wavelength-independent notation. Raman shift wavenumbers are the unit used to do so. The raman shift of a photon is simply

$$\Delta\nu = \nu_0 - \nu_s \quad (17)$$

where ν_0 is the frequency of the incident light and ν_s is the frequency of the scattered light.

This way we can express raman shifts without involving laser wavelengths knowing well that raman spectra will show in the same wavenumber regardless of the laser used.

2 Methodology

2.1 Calibration of the Monochromator

First the monochromator's transmission curves for horizontally and vertically polarized light were looked after. To do so, a Quartz Tungsten-Halogen Lamp was used. This lamp features a stable output and broadband, incoherent illumination in the visible and near-IR portions of the spectrum.

A polarizer was placed at the input of the monochromator as show in figure 3. For further analysis, it is important that both transmission curves are comparable, and hence measured with the same set parameters in the experiment (or somehow made compatible through data processing). After tweaking the parameters of the coupler and the photomultiplier high voltage supply, the following configuration was found to have enough dynamic range to record both horizontal and vertical without saturating the instrument signals: The setup was first manipulated as to fit the output voltage obtained by horizontally polarized light best, just below the saturation point for the whole spectra. The coupler was set in 10^{-6} , the voltage supply for the photomultiplier to ~ 300 V and the spectra was taken from 400 nm to 800 nm at a speed of 5 nm/s.

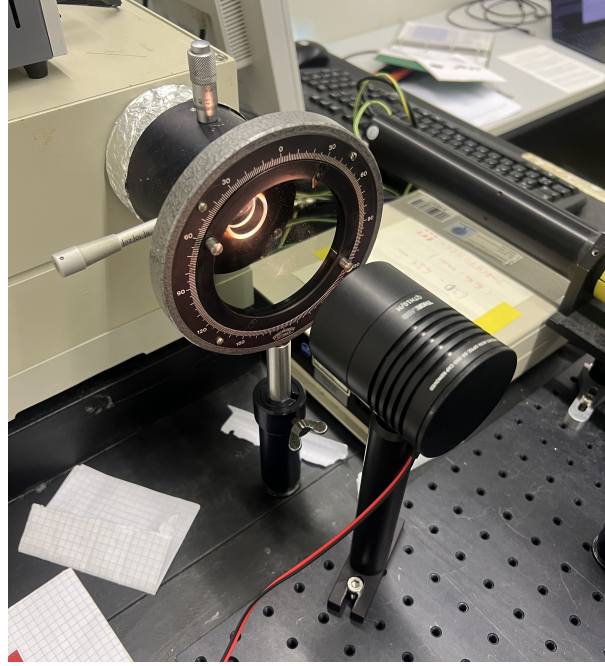


Figure 3: Halogen lamp, polarizer and monochromator input

Then, with the same set of parameters, the polarizer was vertically aligned and once again the spectra was taken from 400 nm to 800 nm at the same speed.

The curve corresponding to horizontal polarization can be seen in figure 4. The curve corresponding to vertical polarization can be seen in figure 5. Both of these figures are raw images extracted from the TDS2001IC oscilloscope screen.

In the experiment, the monochromator grating is adjusted by a servo motor connected to a microcontroller which was programmed to make precise adjustments to the grating. In order to know which voltage corresponds to which wavelength, the arduino's voltage output is also recorder in the oscilloscope to match the data.

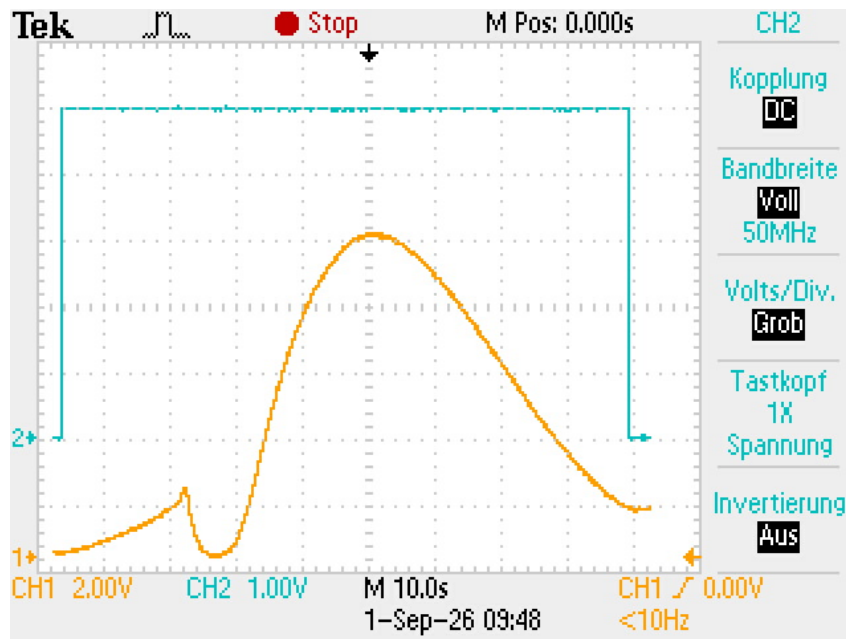


Figure 4: Voltage curve of the photomultiplier for horizontally polarized light (orange) and servo voltage (blue).

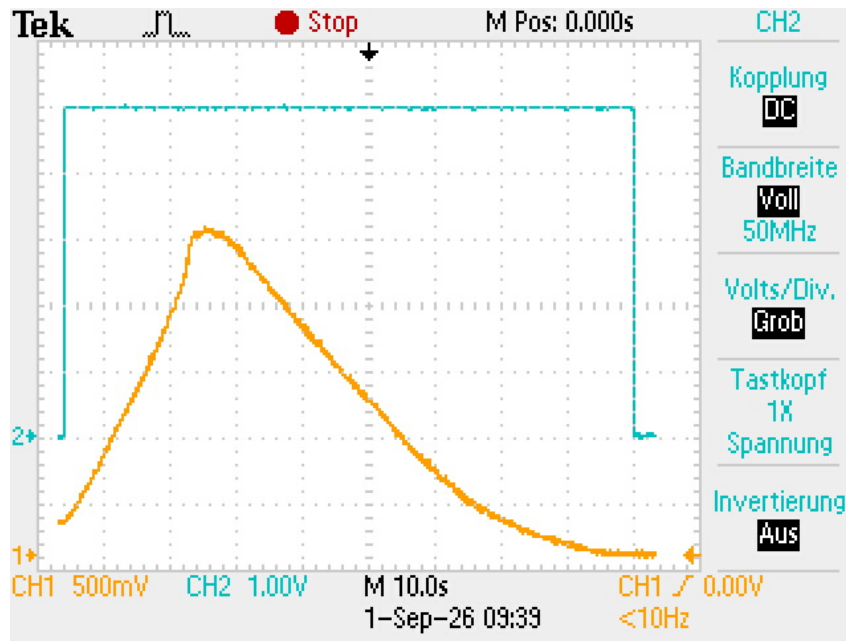


Figure 5: Voltage curve of the photomultiplier for vertically polarized light (orange) and servo voltage (blue).

The data shown in both figures 4 and 5 will be useful to apply instrument response correction (correct the actual raman spectra taking into account the wavelength-dependent intensity of the photocromator).

The Halogen lamp was removed from the array, and the two lenses that form the telescope were placed so that the laser light goes in as focused as possible in the monochromator. The ethanol + salt sample was inserter in the chamber. This sample has a very strong rayleigh reflection, and is thus the ideal sample to align the experiment. The monochromator was swept in between 622 and 642 nm. The Voltage curve can be seen in figure 6. Note that the lock-in mechanism and the chopper are not active yet.

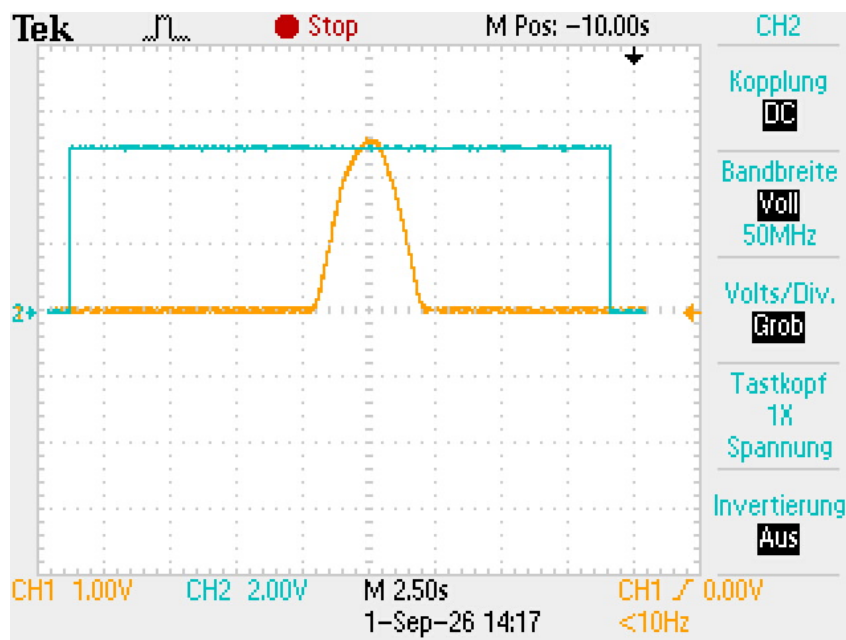


Figure 6: Voltage curve of the 633 nm laser (ethanol +salt cuvette)

Figure 6 shows that our laser indeed has a gaussian profile with a peak presumably at 633 nm.

2.1.1 instrument response correction

Figures 4 and 5 prove that the monochromator has a wavelength-dependent response $R(\lambda)$, thus we can say that approximately

$$I_{measured}(\lambda) = I_{true}(\lambda)R(\lambda) \quad (18)$$

So to correct the measured spectrum we divide by the response of the monochromator.

2.2 Experiment Setup

Having the monochromator response, the rest of the setup can be properly introduced in the experiment. A laser of wavelength of 633 nm goes through a mechanical chopper whose frequency can be adjusted by a control box. The laser then goes through a cage system containing a sample, it is then reflected onto the telescope system. Finally the laser goes inside the photochromator. the photomultiplier then outputs a signal for data analysis. In between the sample and the monochromator, a polarizer is introduced to analyze the change in light polarization after the sample. The laser polarization can also be adjusted by rotating the laser tube.

A diagram of the experiment can be seen in 7.

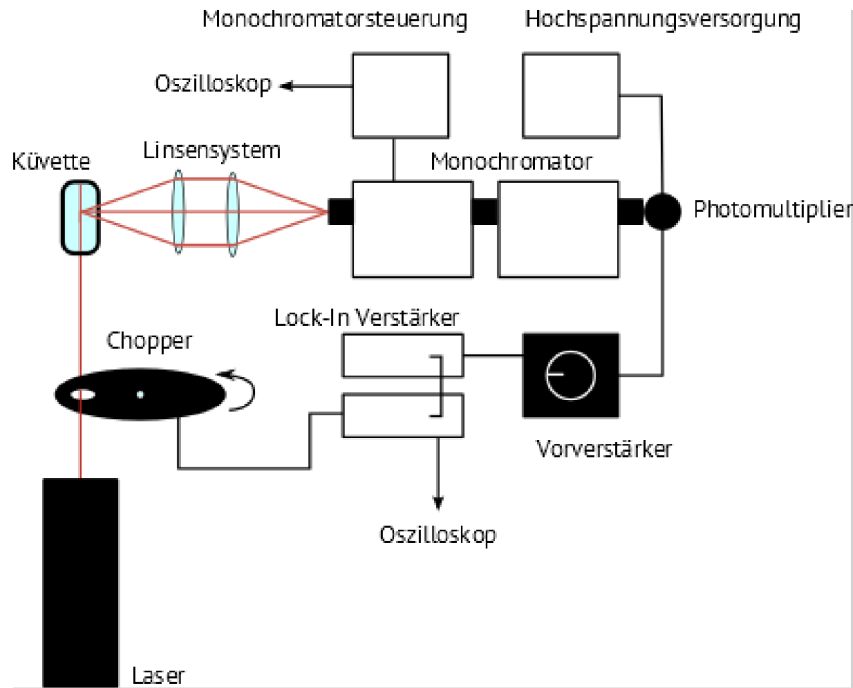


Figure 7: Setup of the experiment. Taken from [5]

After the monochromator, the signal processing electronics take place. Given that the photomultiplier signal is very small, immediately after a preamp is connected to strengthen the signal. Said signal is then fed to a phase sensitive Detector. This detector has as reference the phase-shifted signal of the chopper. A picture of this part of the array is seen in figure 8. These electronics and configuration is explained in detail in section 2.3.1

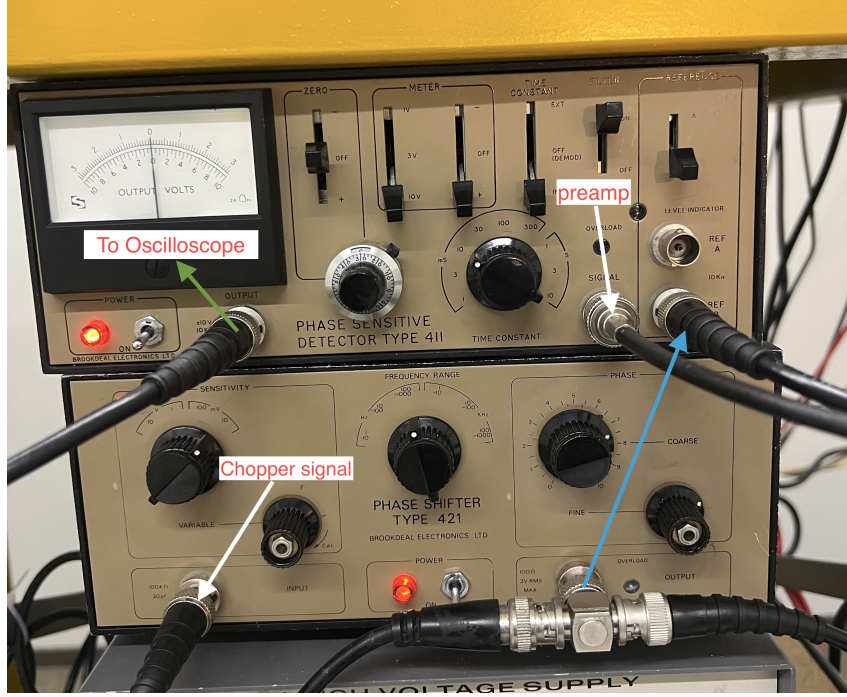


Figure 8: Lock-In amplifier inputs and outputs (knobs are not in final position)

2.3 Lock-In Amplifier

It will be seen in the following sections of this protocol that raman peaks are of very low intensity and can be hard to discern in a regular spectrum; this is because raman scattering is considerably weaker than rayleigh scattering. Often they can be buried in noise and the rayleigh peaks can also bury them if they're overlapping.

To overcome this challenge, we need a way to remove unwanted noise from the spectra. This is where the Lock-In amplifier comes in handy as explained in section 1.2

2.3.1 Lock-in amplifier configuration

The optical chopper was set to an arbitrary frequency (around 130 Hz). The chopper driver generates an AC output that was forwarded to our Phase shifter. The output of the phase shifter is then taken as a reference signal in the phase sensitive detector (PSD). Furthermore, the chopper frequency can be calculated with the AC output signal. The output of the current amplifier (whose input is the photomultiplier) is then taken as the input signal on the PSD. Refer to figure 8 for a visual image of the connections.

The fine-tuning of the lock-in mechanism took place. Given that the chopper has a frequency of around 130 Hz, the phase shifter frequency has to be set accordingly to the range 100 – 1000 Hz. Then the phase shift is adjusted either to match closely either 0° or 180°. This is because the photodiode signal at the chopping frequency has the form

$$S(t) = A \cos(\omega t + \phi) \quad (19)$$

while the reference from the chopper is

$$R(t) = \cos(\omega t + \theta) \quad (20)$$

At the PSD, both signals are multiplied and low-pass filtered, giving an expression resembling

$$V(t) = \frac{A}{2} \cos(\phi - \theta) \quad (21)$$

It is evident that the resulting signal has an absolute maxima when the above condition is met. In practice, this is done by hooking the photodiode and the phase-shifted chopper in the two channels of the oscilloscope and adjusting the time scale to distinguish the square oscillating signal as seen in figure 9

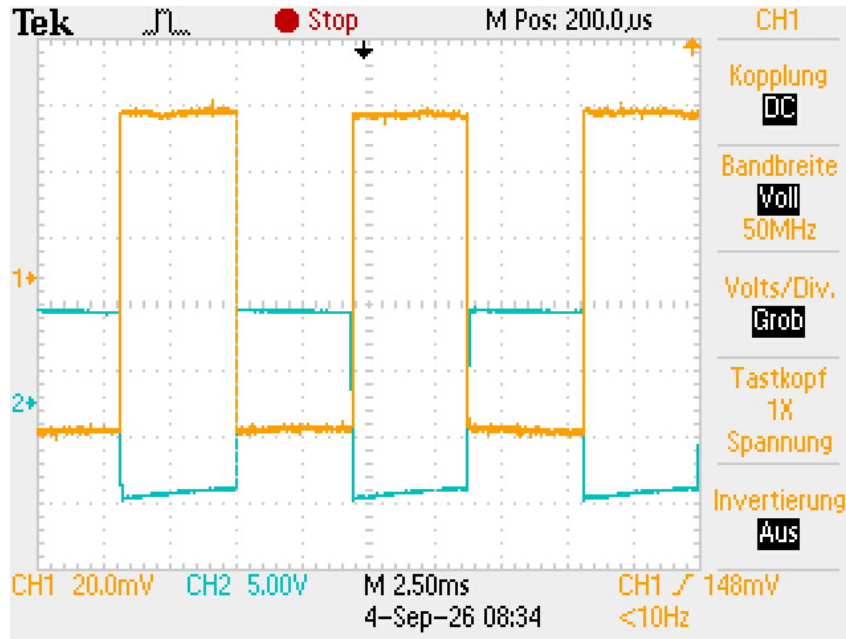


Figure 9: Phase shifted chopper (blue) and photodiode (yellow) on the oscilloscope. They fulfill the locked condition

The time constant α of the PSD was introduced after that. The time constant is the low pass constant which determines how quickly the PSD output is allowed to respond to changes by controlling the bandwidth of its low pass filter.

Simply put, if α is small, the output responds quickly, but more noise gets through. Viceversa, if α is big, output changes slowly, but more noise is rejected.

3 Results

After processing the data by taking into account the servo response, both of the curves seen in figures 4 and 5 are now shown with the wavelength (nm) in figures 10 and 11

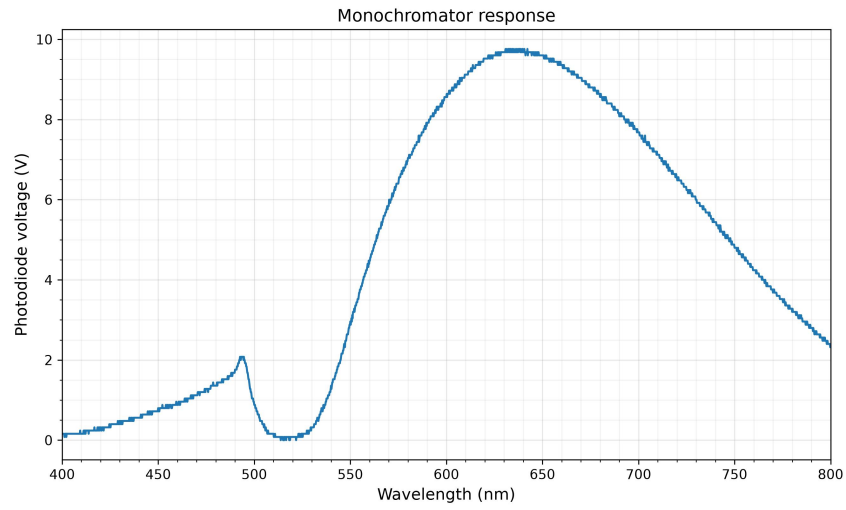


Figure 10: Monochromator response (horizontally polarized light).

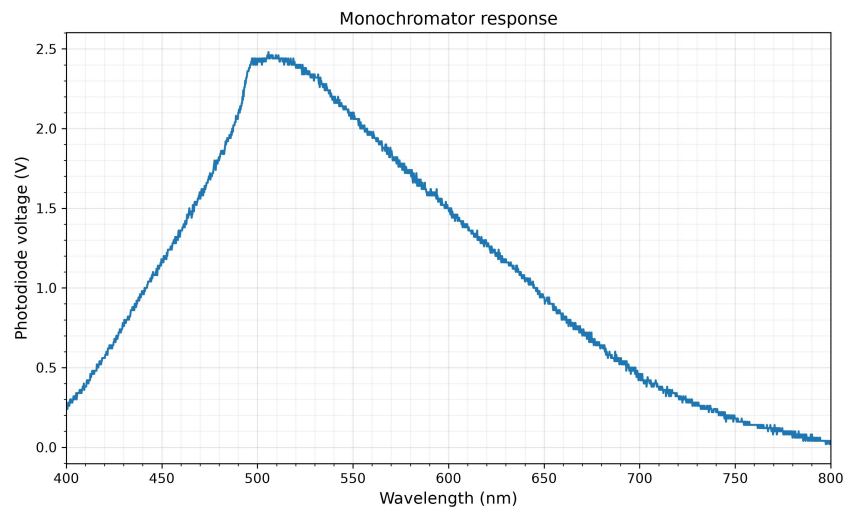


Figure 11: Monochromator response (vertically polarized light).

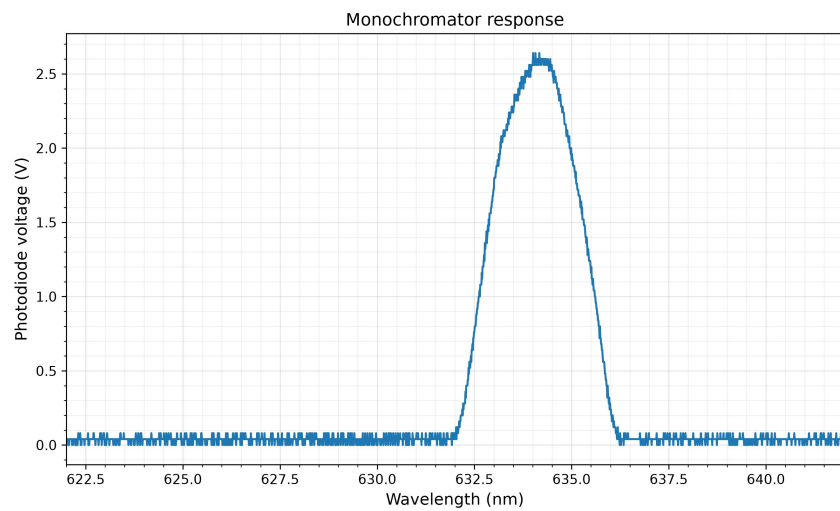


Figure 12: Laser spectra (ethanol + salt)

Take note that all of the next results are corrected to consider the monochromator polarizability/wavelength response as explained in section 2.1.1.

3.1 Spectra and fitting of CCL4

As suggested in section 1.9, all of the results are presented in the appropriate raman shift wavenumber.

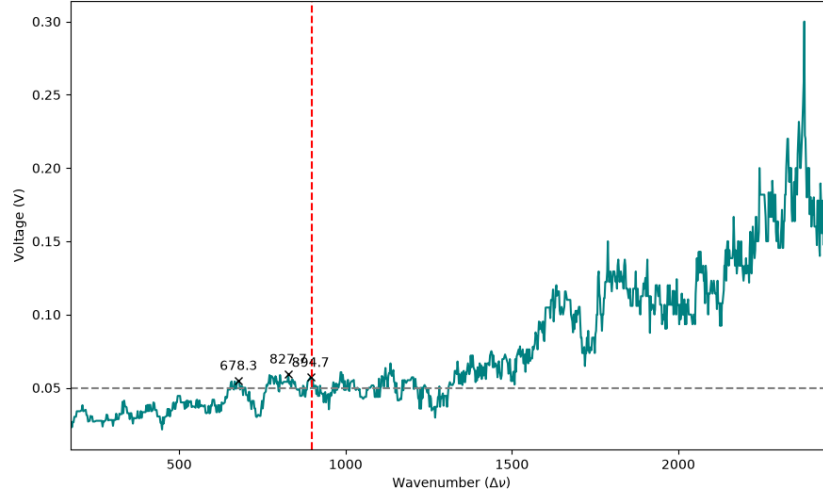


Figure 13: CCL4 spectra with laser at 0° and polarizer at 0°

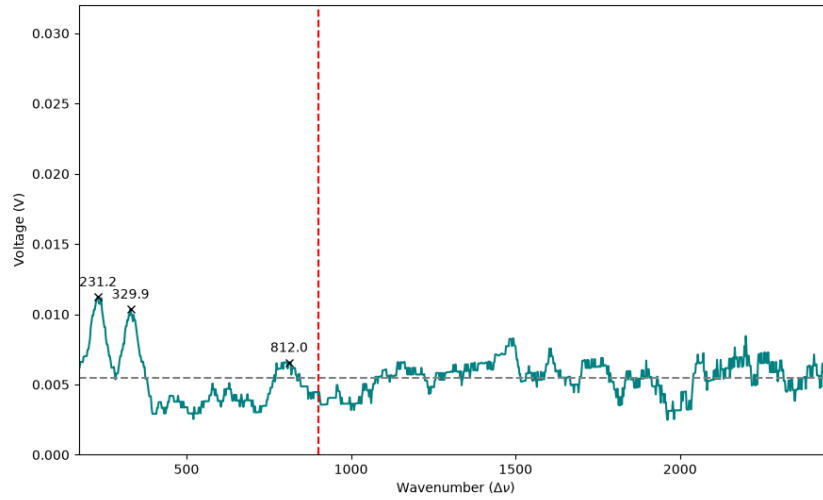


Figure 14: CCL4 spectra with laser at 0° and polarizer at 90°

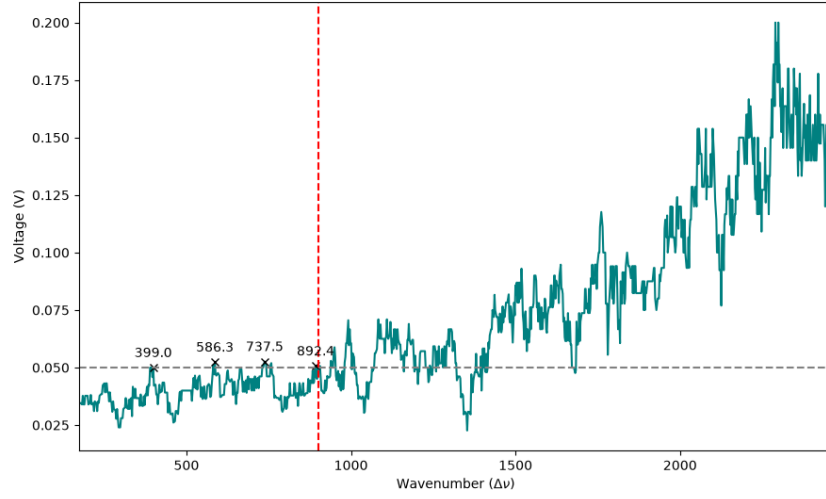


Figure 15: CCL4 spectra with laser at 90° and polarizer at 0°

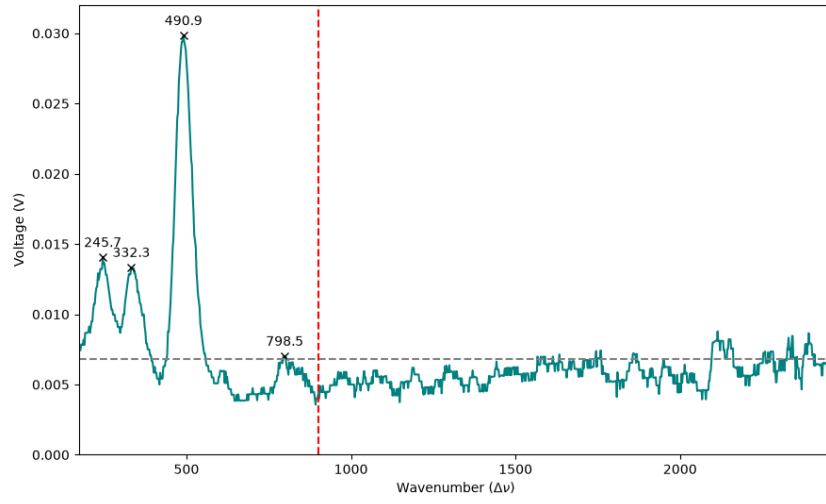


Figure 16: CCL4 spectra with laser at 90° and polarizer at 90°

The following curve fits (figures 17, 35 and 18) were done using the spectra from the measurement shown in figure 16.

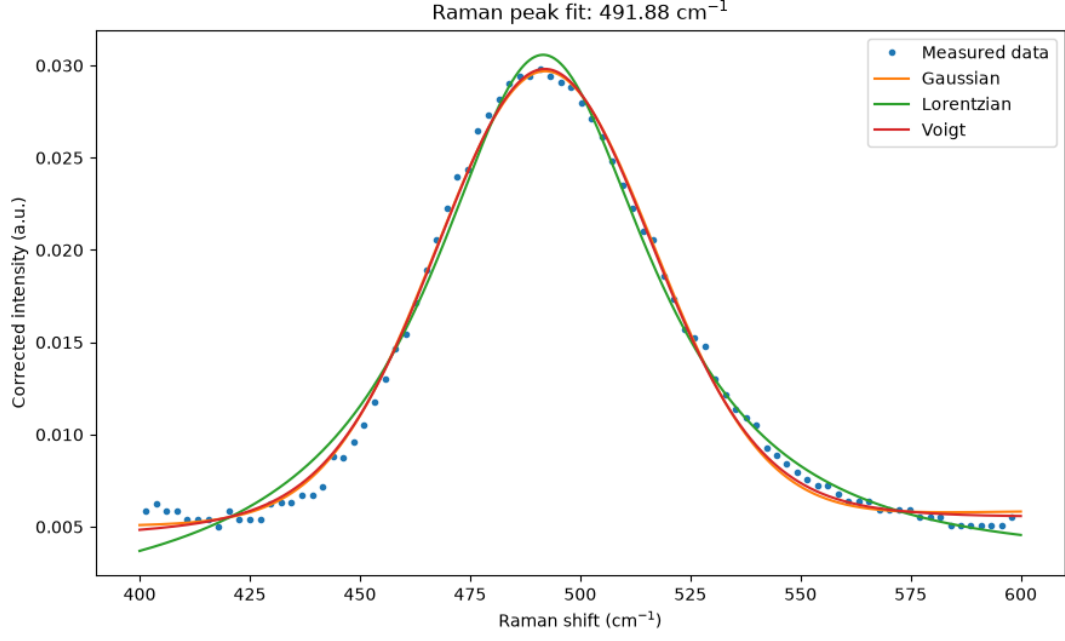


Figure 17: Fit for Raman peak at 491.9 cm^{-1}

Peak	Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm ⁻¹)	R^2
491	Gaussian	492.0 ± 0.2	653.35 ± 0.01	58.2	0.9954
491	Lorentzian	491.4 ± 0.3	653.32 ± 0.01	61.8	0.9902
491	Voigt	491.9 ± 0.2	653.34 ± 0.01	58.8	0.9956

Table 1: Fit parameters for the 491.9 cm^{-1} Raman peak.

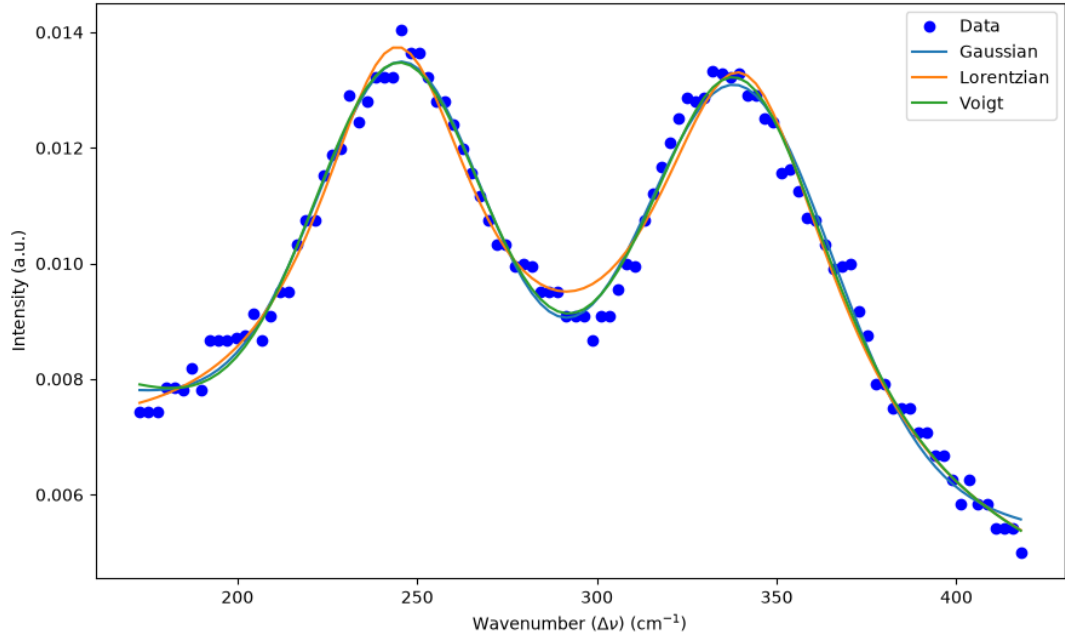


Figure 18: Fit for Raman peak at 491.9 cm^{-1}

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	FWHM (cm ⁻¹)	R^2
245	Gaussian	246.2 ± 0.5	56.0	0.9838
332	Gaussian	339.0 ± 0.4	62.0	0.9838
245	Lorentzian	244.2 ± 0.5	56.1	0.9806
332	Lorentzian	340.5 ± 0.5	64.4	0.9806
245	Voigt	245.0 ± 0.6	52.4	0.9848
332	Voigt	338.7 ± 0.8	67.5	0.9848

Table 2: Fit parameters for the twin peaks 245 cm⁻¹ and 332 cm⁻¹ Raman peak.

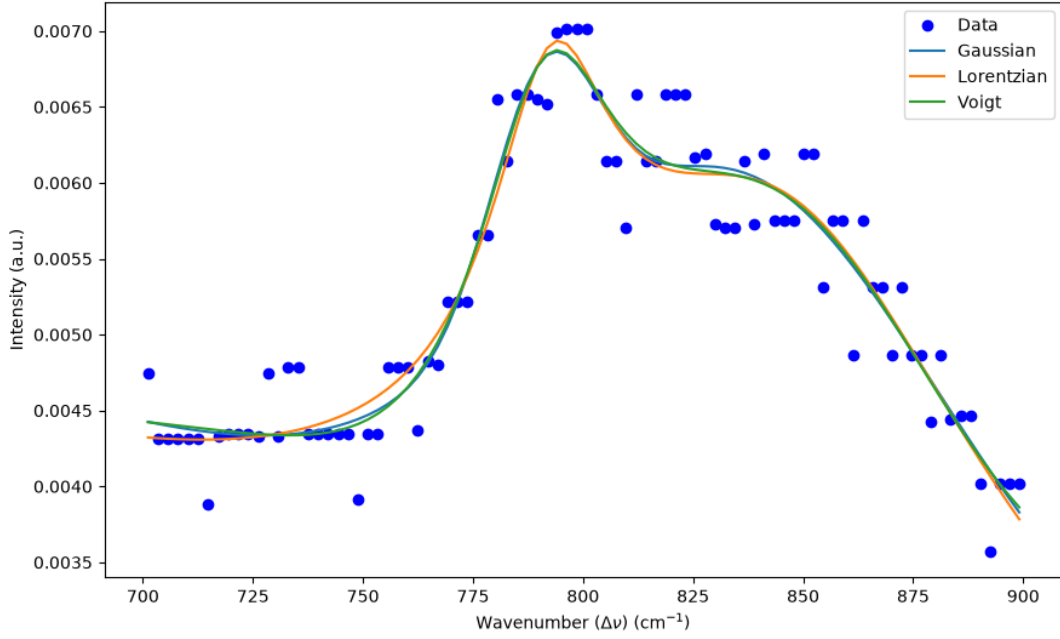


Figure 19: Fit for fermi resonance

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	FWHM (cm ⁻¹)	R^2
780	Gaussian	791.1 ± 1.2	28.3	0.9212
830	Gaussian	835.5 ± 4.5	105.5	0.9212
780	Lorentzian	792.9 ± 1.2	33.6	0.9179
830	Lorentzian	845.1 ± 4.4	121.5	0.9179
780	Voigt	791.6 ± 1.4	41.6	0.9215
830	Voigt	843.5 ± 4.6	65.7	0.9215

Table 3: Fit parameters for the twin peaks 245 cm⁻¹ and 332 cm⁻¹ Raman peak.

3.2 Spectra and fitting of Benzene

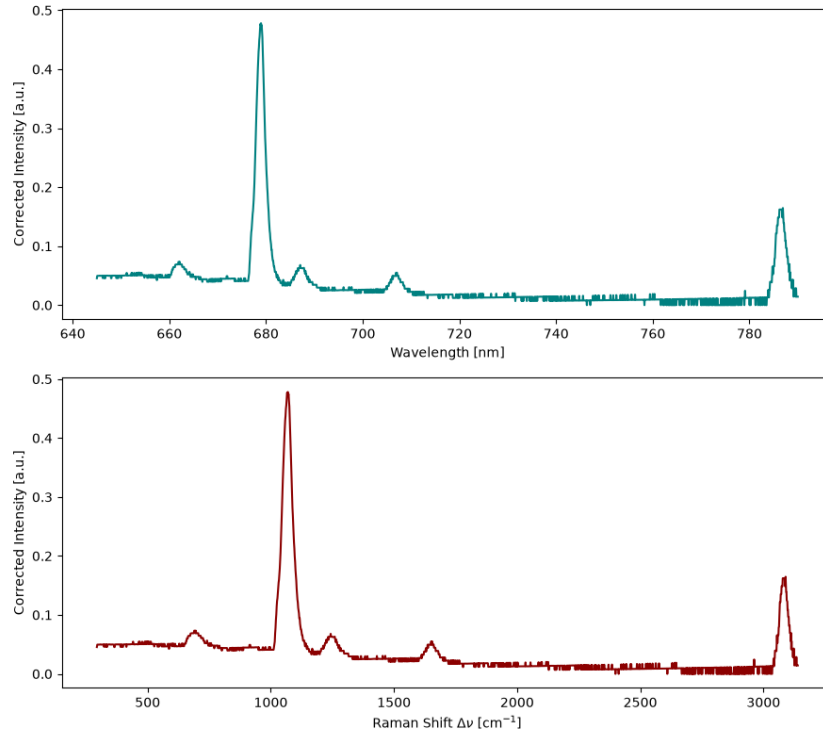


Figure 20

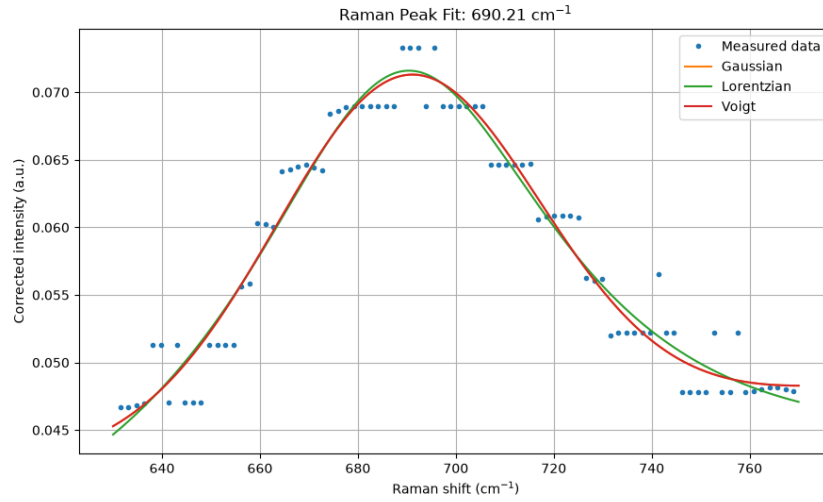


Figure 21

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
688.000000	Gaussian	690.2 ± 0.8	661.92 ± 0.04	63.3	0.9539
688.000000	Lorentzian	689.4 ± 0.9	661.88 ± 0.04	80.6	0.9511
688.000000	Voigt	690.2 ± 0.9	661.92 ± 0.04	63.3	0.9539

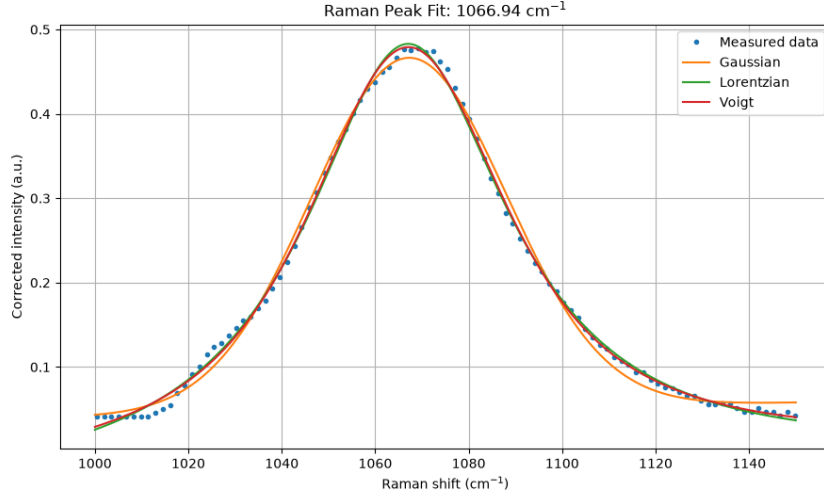


Figure 22

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
1068.000000	Gaussian	1067.2 ± 0.2	678.86 ± 0.01	49.2	0.9944
1068.000000	Lorentzian	1066.9 ± 0.1	678.85 ± 0.01	53.3	0.9978
1068.000000	Voigt	1066.9 ± 0.1	678.85 ± 0.01	52.2	0.9981

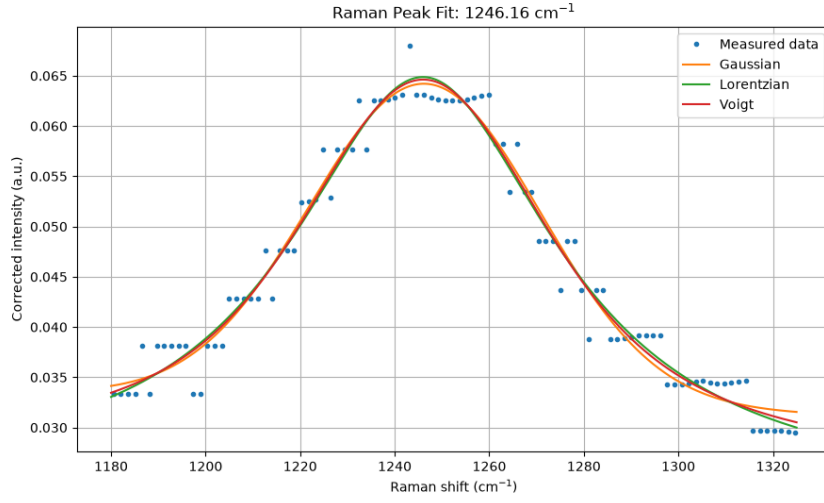


Figure 23

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
1250.000000	Gaussian	1246.4 ± 0.5	687.22 ± 0.03	57.6	0.9692
1250.000000	Lorentzian	1246.1 ± 0.6	687.20 ± 0.03	68.8	0.9703
1250.000000	Voigt	1246.2 ± 0.6	687.21 ± 0.03	64.1	0.9709

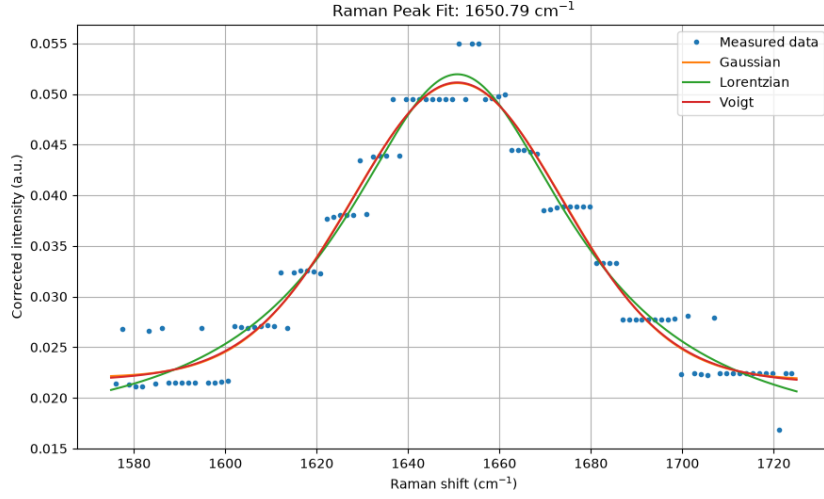


Figure 24

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
1650.000000	Gaussian	1650.8 ± 0.5	706.86 ± 0.03	54.2	0.9605
1650.000000	Lorentzian	1650.8 ± 0.6	706.86 ± 0.03	61.2	0.9587
1650.000000	Voigt	1650.8 ± 0.5	706.86 ± 0.03	54.9	0.9605

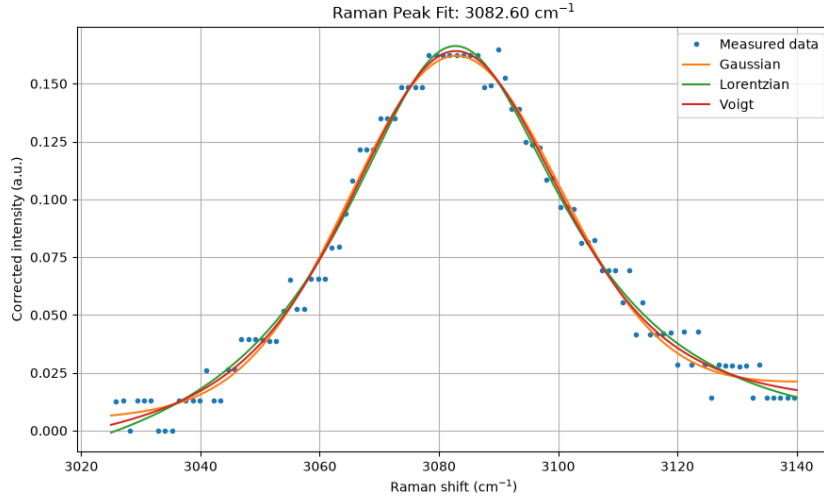


Figure 25

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
3080.000000	Gaussian	3082.6 ± 0.3	786.46 ± 0.02	41.4	0.9847
3080.000000	Lorentzian	3082.6 ± 0.3	786.46 ± 0.02	47.0	0.9846
3080.000000	Voigt	3082.6 ± 0.3	786.46 ± 0.02	44.0	0.9857

3.2.1 Benzene antistokes

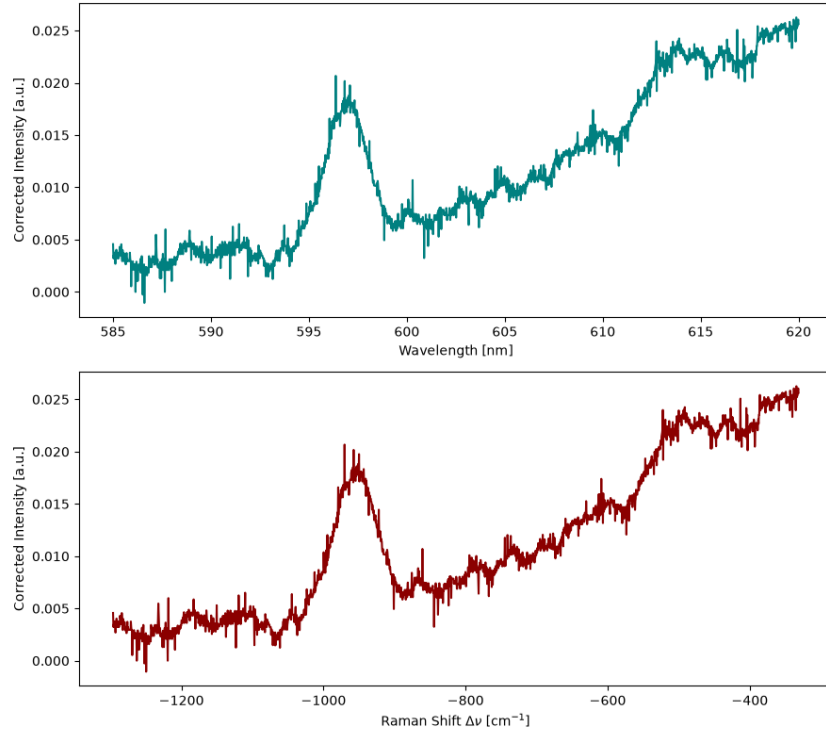


Figure 26

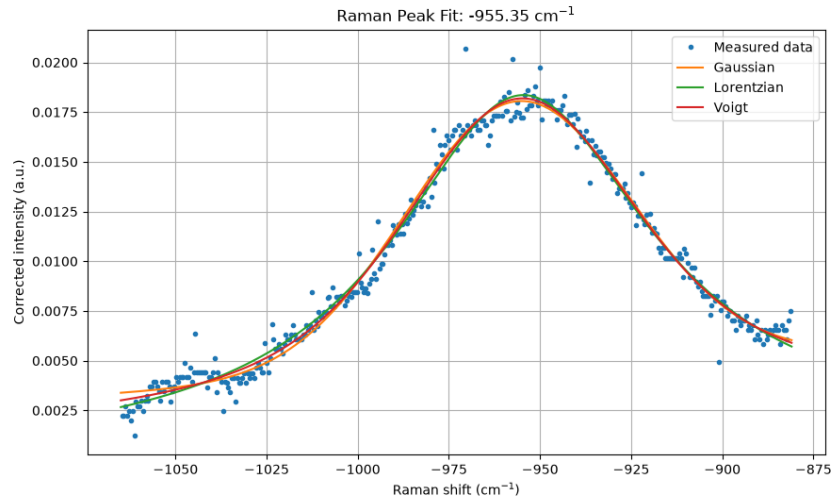


Figure 27

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
-950.000000	Gaussian	-955.9 ± 0.3	596.89 ± 0.01	73.0	0.9826
-950.000000	Lorentzian	-954.8 ± 0.3	596.92 ± 0.01	87.1	0.9824
-950.000000	Voigt	-955.4 ± 0.3	596.90 ± 0.01	79.0	0.9835

3.3 Unknown sample Spectra

Figure 28 shows the spectra recorded for the sample whose content was not known in advance.

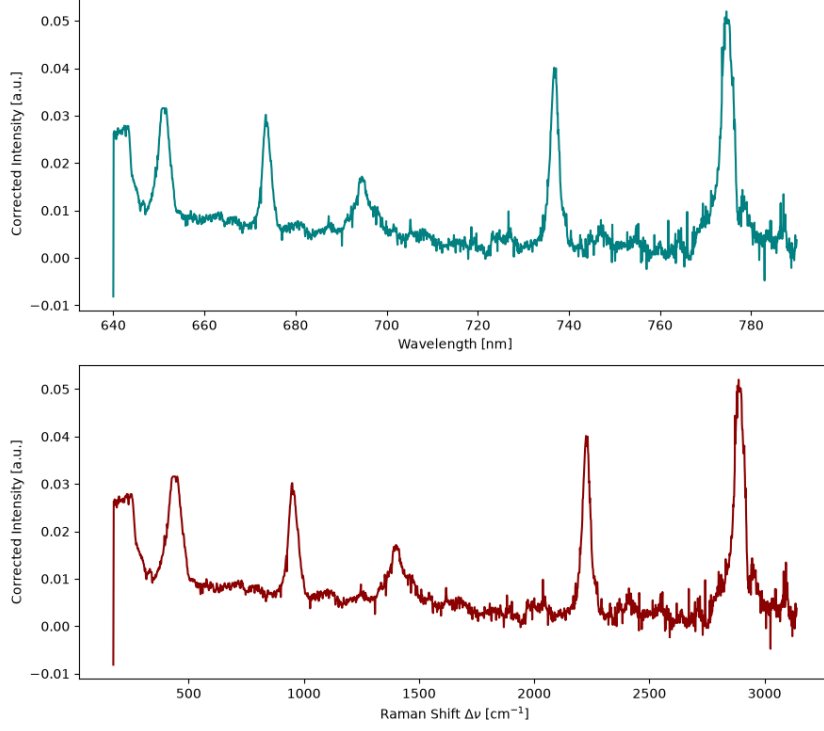


Figure 28: Spectra of *unknown sample 1*.

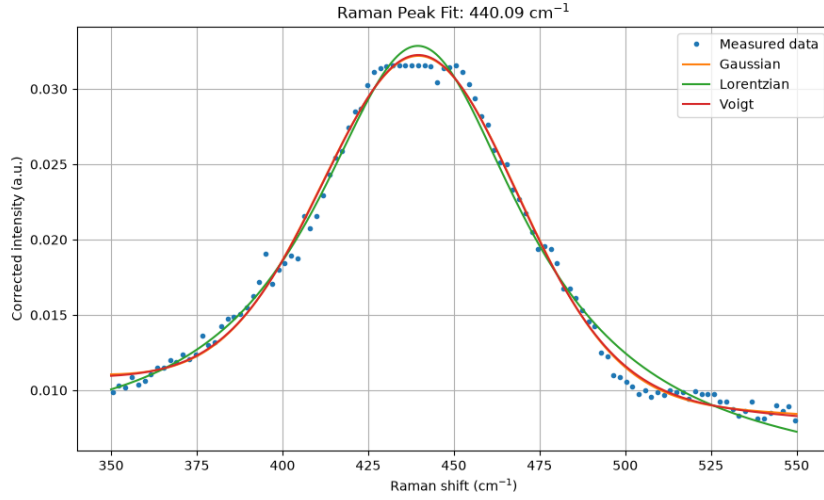


Figure 29: Spectra of *peak at 450 cm⁻¹*.

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
440.000000	Gaussian	440.1 ± 0.3	651.14 ± 0.01	67.2	0.9929
440.000000	Lorentzian	439.7 ± 0.4	651.12 ± 0.02	74.9	0.9889
440.000000	Voigt	440.1 ± 0.3	651.14 ± 0.01	67.9	0.9930

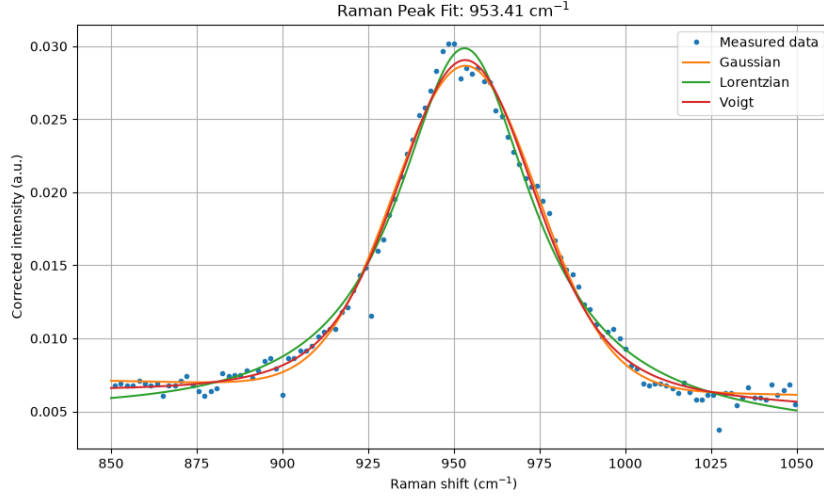


Figure 30: Fit for *peak at 950 cm⁻¹*.

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
950.000000	Gaussian	953.5 ± 0.3	673.66 ± 0.01	49.2	0.9876
950.000000	Lorentzian	953.2 ± 0.3	673.64 ± 0.01	48.5	0.9852
950.000000	Voigt	953.4 ± 0.2	673.66 ± 0.01	49.0	0.9889

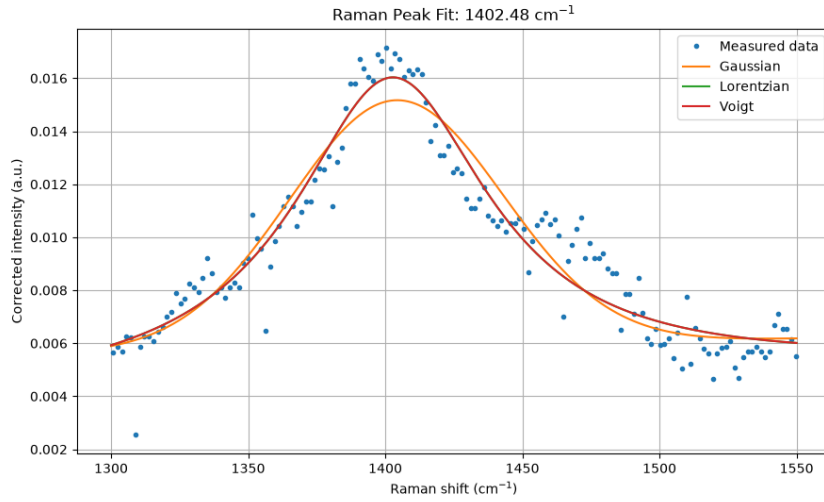


Figure 31: Fit for *peak at 1410 cm⁻¹*.

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
1410.000000	Gaussian	1404.0 ± 1.3	694.75 ± 0.06	91.8	0.8903
1410.000000	Lorentzian	1402.5 ± 1.0	694.67 ± 0.05	86.8	0.9146
1410.000000	Voigt	1402.5 ± 1.0	694.67 ± 0.05	86.8	0.9146

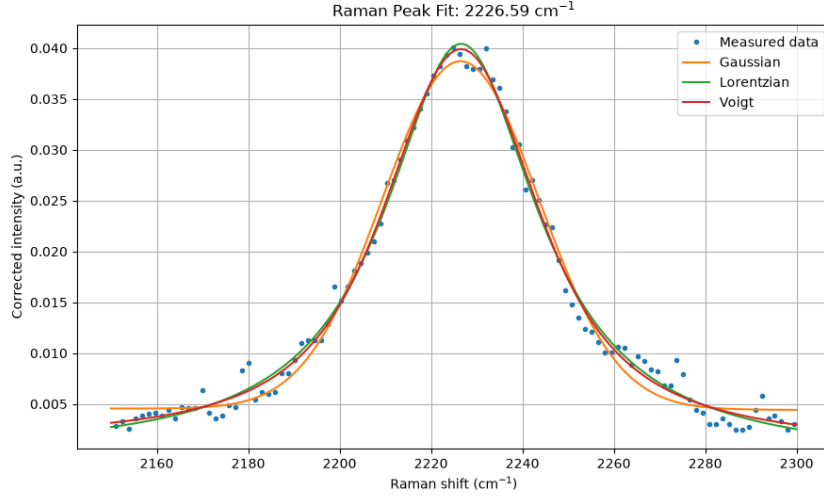


Figure 32: Fit for *peak at 2226 cm⁻¹*.

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
2226.000000	Gaussian	2226.5 ± 0.2	736.85 ± 0.01	40.2	0.9857
2226.000000	Lorentzian	2226.6 ± 0.2	736.86 ± 0.01	40.9	0.9905
2226.000000	Voigt	2226.6 ± 0.2	736.85 ± 0.01	40.5	0.9912

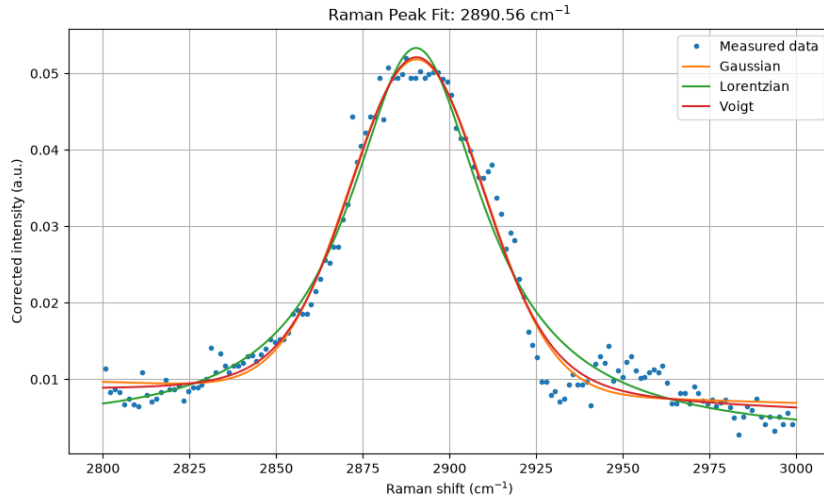


Figure 33: Fit for *peak at 2888 cm⁻¹*.

Peak	Fit	$\Delta\nu$ (cm ⁻¹)	λ (nm)	FWHM (cm ⁻¹)	R^2
2888.000000	Gaussian	2890.6 ± 0.3	774.76 ± 0.02	46.0	0.9718
2888.000000	Lorentzian	2890.3 ± 0.4	774.75 ± 0.02	47.2	0.9638
2888.000000	Voigt	2890.6 ± 0.3	774.76 ± 0.02	46.3	0.9727

3.4 Ethanol concentration of an unknown sample

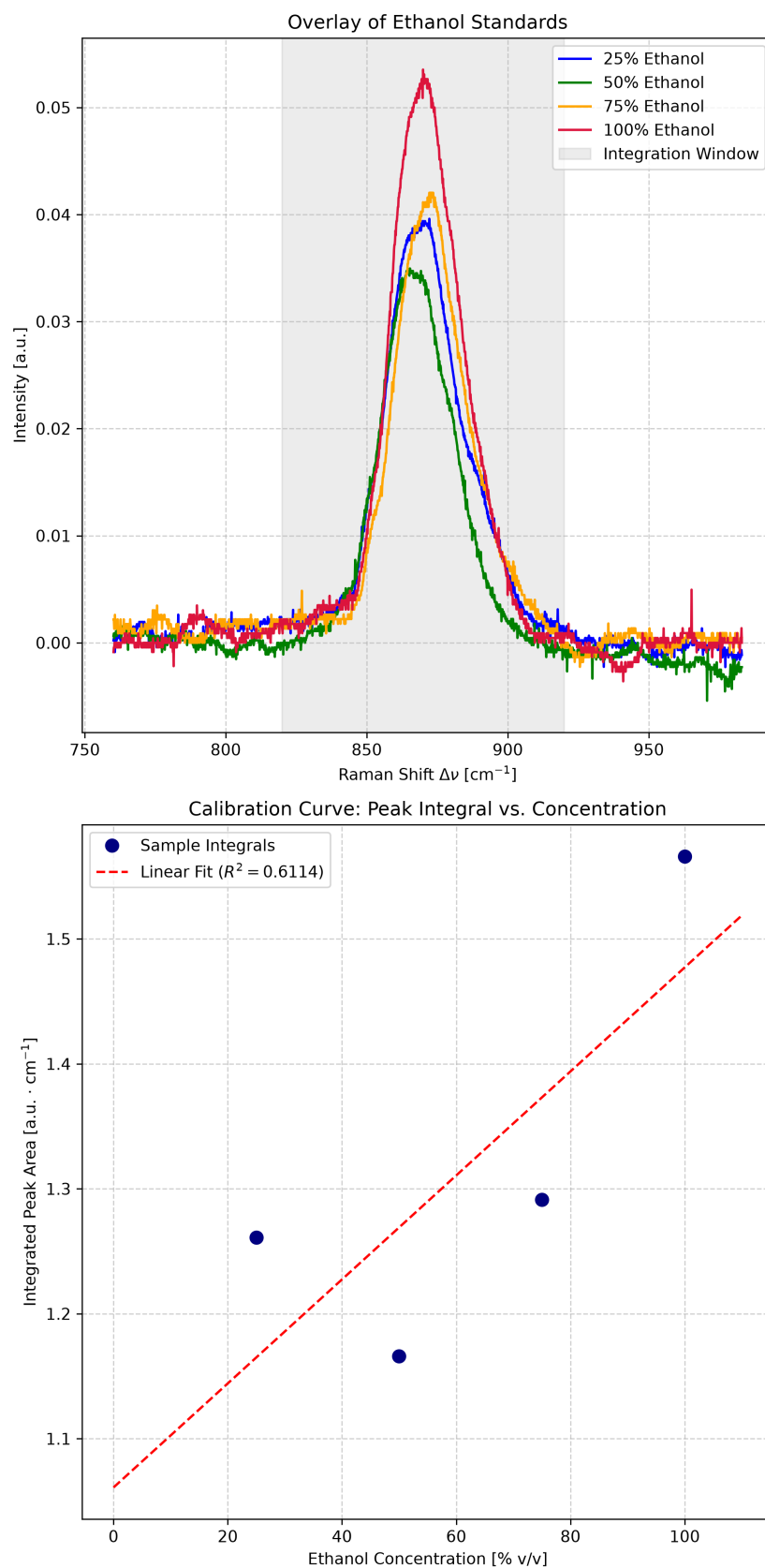


Figure 34

Table 4: Ethanol Calibration Fit Results

Concentration (%)	Integrated Area
25	1.2609
50	1.1661
75	1.2912
100	1.5661
Linear Fit Statistics	
Equation: $\text{Area} = 0.0042 \times C + 1.0610$	
$R^2 = 0.61135$	

3.5 Fit of relevant peaks

3.5.1 CCL4

3.5.2 Benzene

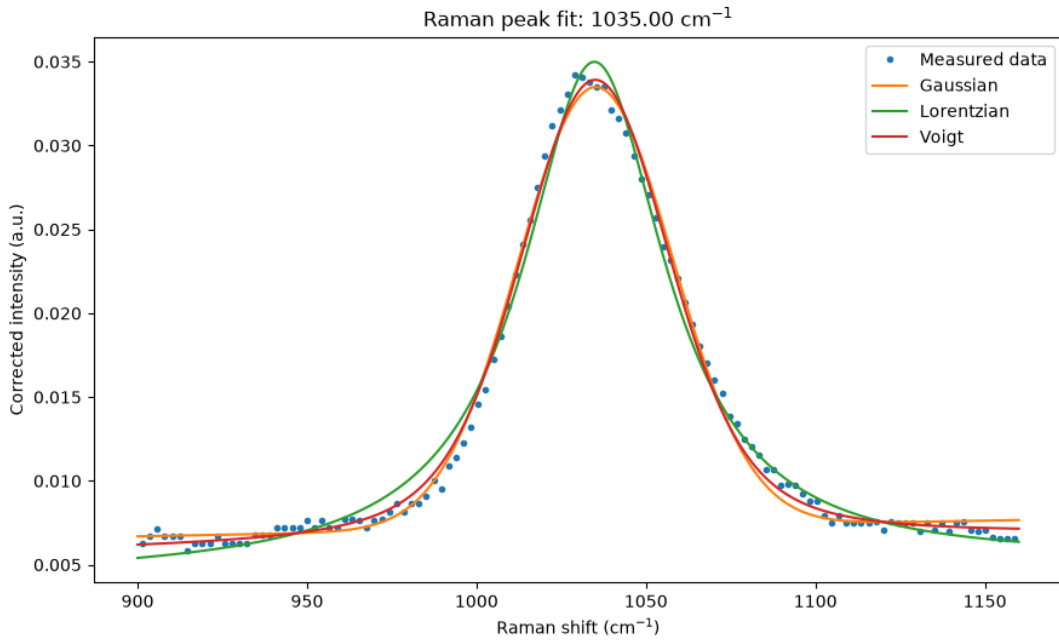


Figure 35: Fit for Raman peak at 1035 cm^{-1}

Peak	Fit	$\Delta\nu \text{ (cm}^{-1}\text{)}$	$\lambda \text{ (nm)}$	FWHM (cm^{-1})	R^2
1028	Gaussian	1035.1 ± 0.2	677.38 ± 0.01	53.8	0.9934
1028	Lorentzian	1034.7 ± 0.3	677.37 ± 0.01	51.4	0.9892
1028	Voigt	1035.0 ± 0.2	677.38 ± 0.01	53.2	0.9950

Table 5: Fit parameters for the Benzene 1035 cm^{-1} Raman peak.

4 Discussion

4.1 CCl₄

Taking a closer look at the CCL4 spectra above, notice that not all of the laser/analyzer configurations yield expected results. If we consider exclusively the sample and the relative orientation of the incident polarization and analyzer, we would expect a simmetrical behavior of the experiment configurations. CCL4 has totally symmetric A modes and non-symmetric F modes.

If we simplify the experiment to two orthogonal polarization directions x and y , we can think of the experiment yielding symmetric results as in table 6. This is however not the case. In reality, the experiment doesn't yield the expected symmetry but instead the ones shown in table 7.

Give that CCL4 is an isotropic sample, there is no molecular reason for this to dissapear while the other survives. The most likely explanation is that the optical system has an additional polarization-dependent transmission/collection effect. As already mentioned, the monochromator used in the experiment has a different response to the orientation of light. This combined with an improper optical alignent is one likely explanation as to why we cannot resolve the lines with the analyzer at the 0° degrees configuration.

Table 6: Idealized polarization dependence of Raman scattering in an isotropic CCl₄ sample. The A_1 mode produces polarized scattering, while the F_2 modes produce depolarized scattering.

Laser polarization	Analyzer polarization	
	0°	90°
0°	A_1	F_2
90°	F_2	A_1

Table 7: Results of the experiment

Laser polarization	Analyzer polarization	
	0°	90°
0°	noise	F_2
90°	noise	A_1

To discuss the number of frencuency shifts in our spectra, it's relevant to explain the molecular structure of CCL4 and its possible vibrational modes. A very didactic animation can be found on [6]([click here](#)). A diagram is also sketched for those who are not reading this report on an e-device.

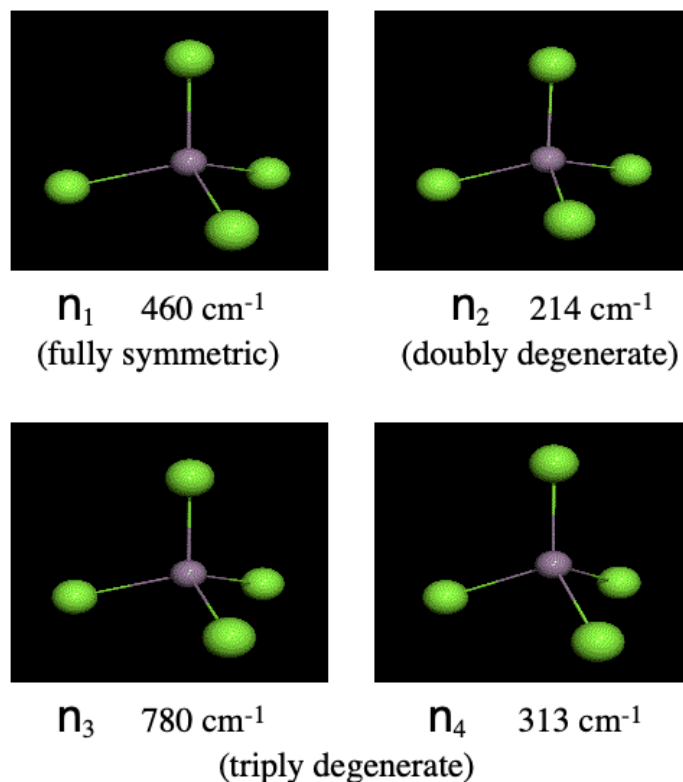


Figure 36: Vibrational modes of Carbon tetrachloride. Adapted from [6]

Each of these vibrational modes influences the amount of lines seen in the Raman spectra.

The vibrational modes of CCl_4 can be classified according to the symmetry of the molecule, resulting in four distinct fundamental modes: $\nu_1(A_1)$, $\nu_2(E)$, $\nu_3(F_2)$, and $\nu_4(F_2)$. Consequently, four distinct fundamental vibrational frequencies are expected in the Raman spectrum, and are indeed seen in figure 16, although the peak corresponding to ν_3 is barely perceptible in our measurements.

Peak ν_3 corresponds to a Fermi resonance. Because of this, instead of observing a single peak, we see a double peak profile in figure 37. This is why the fit shown in figure 19 accounts for a two peak fit, even if in our data this distinction is not perceived.

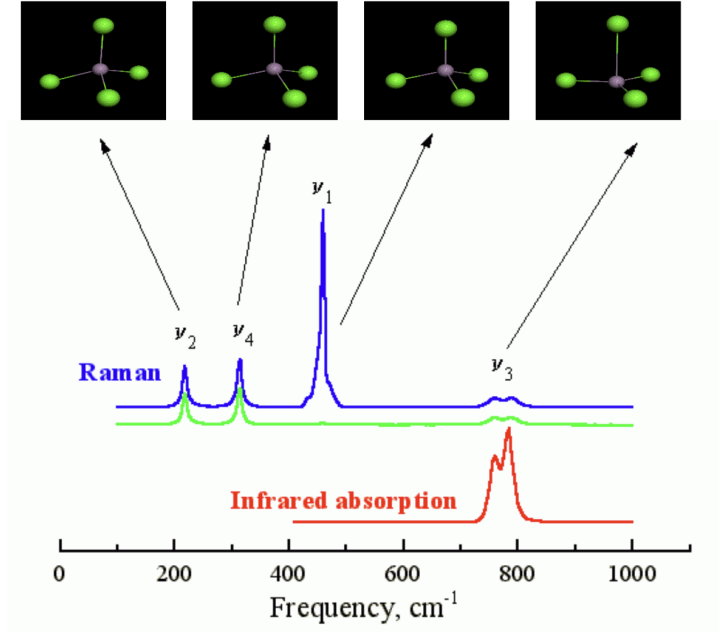


Figure 37: Raman peaks corresponding to vibrational modes. Adapted from [6]

For tables 8 and 10, the following notation is useful: $C = 3 \sim 6\text{cm}^{-1}$, $E = 15 \sim 30\text{cm}^{-1}$,

As it is shown on the mentioned tables, our measurements are off by a few dozens of wavenumbers. Most likely the reference between the software/monochromator was not properly adjusted before the measurements. and hence a displacement of wavenumbers occurred in our data.

Table 8: Literature and measured Raman bands of carbon tetrachloride (CCl_4).

Symmetry	Mode	NIST ν 1/cm	Uncertainty 1/cm	Measured ν 1/cm	Deviation 1/cm
a_1	Symmetric stretch	458.7	C	490.9	32.2
e	Degenerate deformation	217.0	C	245.7	28.7
f_2	Degenerate stretch	790.4	E	798.5	8.1
f_2	Degenerate stretch	761.7	E	—	—
f_2	Degenerate deformation	313.5	C	332.3	18.8

4.1.1 polarization of CCL4 modes

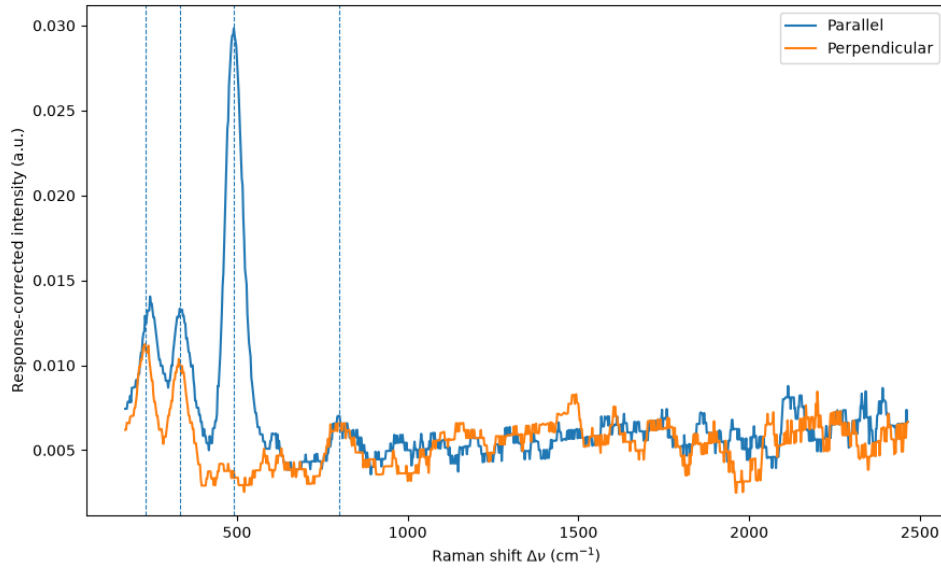


Figure 38: Horizontal and Perpendicular measurements superimposed

$\Delta\nu$ (cm ⁻¹)	$I_{\parallel}$ (a.u.)	$I_{\perp}$ (a.u.)	ρ
231	0.0132	0.0112	0.851
332	0.0133	0.0104	0.778
490	0.0298	0.00378	0.127
798	0.00702	0.00658	0.937

Table 9: polarization ratios for the vibrational modes of CCL4

Table 9 shows the calculated polarization ratios from the measurements taken. Our measurements are in agreement with the literature reported in [7]. The only strongly polarized peak corresponds to the vibrational mode $\nu(A_1)$, since it corresponds to a diagonal entry in the raman polarization tensor of CCL4. The rest of the peaks ν_2 , ν_3 and ν_3 correspond to non-diagonal entries in the tensor, and are thus depolarized as seen in both figure 38 and table 9

4.2 Benzene

4.2.1 peaks

Tables 8 and 10 show the reported raman shifts according to [7] against the measured shifts in the experiment.

For Benzene, we were able to record only the most prominent peak, located at 991.6 cm^{-1} . The rest of the peaks are labeled in the literature (see [7]) as medium or low intensity peaks. As mentioned, our alignment was not providing a precise input scattered beam to the monochromator, so little intensity for the raman peaks of benzene was obtained.

Table 10: Literature and measured Raman bands of Benzene (C₆H₆)

Symmetry	Mode	NIST ν 1/cm	Uncertainty 1/cm	Measured ν 1/cm	Deviation 1/cm
a_{1g}	CH stretch	3061.9	C	—	—
a_{1g}	Ring stretch	991.6	C	1028.9	37.3
a_{2g}	CH bend	1326.0	E	—	—
e_{1g}	CH bend	848.9	C	—	—
e_{2g}	CH stretch	3046.8	C	—	—
e_{2g}	Ring stretch	1606.4	E	—	—
e_{2g}	Ring stretch	1584.6	E	—	—
e_{2g}	CH bend	1178.0	C	—	—
e_{2g}	Ring deformation	605.6	C	—	—

4.2.2 peak polarization

By observing figures 14 and 16, it is evident that the peak we measured at 490.9 cm^{-1} corresponds to strongly polarised Raman scattering. Approximating from the values recorded

$$\rho = \frac{I_{\perp}}{I_{\parallel}} \approx \frac{0.004}{0.030} = 0.13 \quad (22)$$

comparing with what was discussed in section 1.5, is is indeed strongly polarized.

The rest of the peaks recorded for CCL4 are not strongly polarized, and from the graphs it can be estimated that $\rho \geq \frac{3}{4}$ for these.

Similarly, the only peak measured for benzene is only really visible on figure ?? which indicates strong polarization without any calculation needed.

[7] also reports the same polarization degrees as in our experiment

4.3 unknown samples

To find out the content of the unknown sample, the [Search for Species Data by Vibrational Energy Value](#) of the NIST was used. After introducing the wavenumbers recorded in the experiment, a comparison between many molecules was done. given that we’re lacking the polarization of the peaks, it’s harder to pinpoint the actual content of the cuvette. Many molecules and peaks were looked after and a lot of the search filters were applied. Ultimately, we suspect that our spectra is heavily shifted, and we couldn’t identify either of the two samples.

If we dismiss the assumption that the spectra is shifted, then the spectra for sample 1 (figure 28) contains two of the strong peaks of Borine carbonyl $\text{CH}_3^{11}\text{BO}$, albeit slightly shifted from where they should be (2380 cm^{-1} recorded at 2384 cm^{-1} and 2169 cm^{-1} recorded at 2145 cm^{-1}). However, the second value is still far enough to classify it as such.

Regarding the second sample, it was originally thought the peaks looked similar to bromoform raman peaks, but upon further inspection, the relative intensities between peaks do not match.

We conclude this discussion suggesting that narrowing down the sample would’ve been easier if polarization data had been recorded during the experiment, as well as a more careful alignment of the analog/software calibration of the wavenumber.

4.4 Determination of the ethanol concentration of an unknown sample

Upon further inspection, we notice that all of the ethanol samples (Figures ?? to ??) show no clear correlation between peak intensities or position. It is unclear how to proceed to make this four spectra a ruler for ethanol concentration in unknown samples. Adding the fact that most likely our unknown spectra is heavily shifted, we opted to avoid making poorly-founded calculations of ethanol concentration.

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